

Math 526, S12

Quiz 5

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Which of the following subsets of $\mathbb{R}^3$ are actually subspaces? (no explanation needed)

(a) The plane of vectors (b_1, b_2, b_3) with $b_1 = b_2$.

Yes

(b) The plane of vectors (b_1, b_2, b_3) with $b_1 = 1$.

No

(c) All linear combinations of $\mathbf{v} = (1, 4, 0)$ and $\mathbf{w} = (2, 2, 2)$.

Yes

(d) All vectors that satisfy $b_1 + b_2 + b_3 = 0$.

Yes

Problem 2. (4 points) Find the factorization of the symmetric matrix $A = LDL^T$ with

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}.$$

Solution: $A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix} \xrightarrow{e_{21} = 3} \begin{bmatrix} 1 & 3 \\ 0 & -7 \end{bmatrix}$

$$A = \underset{L}{\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}} \underset{U}{\begin{bmatrix} 1 & 3 \\ 0 & -7 \end{bmatrix}}$$

$$= \underset{L}{\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}} \underset{D}{\begin{bmatrix} 1 & 0 \\ 0 & -7 \end{bmatrix}} \underset{L^T}{\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}}$$